

NINGKE SUN

Boston University, Department of Mathematics and Statistics
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RESEARCH INTERESTS

Number Theory and Arithmetic Geometry. More specifically Elliptic Curves, Automorphic Forms, algebraic groups and Murmuration Density

EDUCATION

Boston University, Boston, MA Sept 2023-May 2027 (Expected)
B.A. in Mathematics, Pure Mathematics Track; Minor in Computer Science GPA: 3.79/4.00
Honors in the Major
Senior Thesis Advisor: Prof. Alexander Bertoloni Meli

RESEARCH

- **Murmuration of ${}_3F_2$ hypergeometric families(In Progress)** 2026
Joint with Daniel Kim, Jamie Le, and Kosuke Sato *Advisor: Tian An Wong*
We extend the approach of Sawin–Sutherland (2025) for elliptic curves ordered by naive height to the symmetric squares of the Legendre family, which are related to ${}_3F_2$ hypergeometric functions.
- **The Crossing Number of the 2-Twist Spun Figure-8 is 8 (In Progress)** 2026
Joint with Monica Sing, Vakul Raghavan and Xinyue Gu
We develop new bounds for tri-plane crossing numbers of knotted surfaces, including an exact computation for the two-twist spun figure-eight knot. We also obtain the first nontrivial lower bound for ribbon 2-knots.

SELECTED EXPOSITORY NOTES

- **Serre’s Open Image Theorem** 2026
An expository account of Serre’s Open Image Theorem for elliptic curves.
- **A Brief Introduction to Morse Theory [\[PDF\]](#)** 2025
Joint with Huan Zhang
An expository introduction to Morse theory and its relationship with CW complexes.

MATHEMATICS EXPERIENCE

- **Reading Course, Boston University** Fall 2026
Algebraic Number Theory *Supervisor: Prof. David Rohrlich*
Principal Text: Algebraic Theory of Numbers by Pierre Samuel
- **Reading Course, Boston University** Spring 2026
Representation Theory *Supervisor: Prof. Alexander Bertoloni Meli*
Principal Text: Linear Representation of Finite Groups by Jean-Pierre Serre, Translated by Leonard L. Scott
- **Briding the LMFDB and Lean(Planned)** January 2027
- **PolyMathJR Undergraduate Research Program** Summer 2026
Topic: Murmuration of Hypergeometric Families *Supervisor: Prof. Tian An Wong*
- **PCMI 2026 Undergraduate Summer School** (Institute for Advanced Study) Jul. 2026
Research Theme: Knotted Surfaces in Four-Manifolds
- **Connecticut Summer School in Number Theory** Jun. 2026
- **PolyMathJR Undergraduate Research Program** Summer 2025
Topic: Ribbon Links *Supervisor: Alex Zupan*

- **Directed Reading Program, Boston University**
Harmonic Analysis
Principal Text: A First Course in Harmonic Analysis by Anton Deitmar

Spring 2025
Supervisor: Jacksyn Bakeberg

TALKS

- *Murmuration of ${}_3F_2$ Hypergeometric Families* Jan. 2027
Joint Mathematics Meetings 2027 (Planned)
- *On the Tri-plane Crossing Number of Knotted Surface* Jan. 2027
Joint Mathematics Meeting 2027 (Planned)
- *Tri-plane diagrams and Minimum Crossing Number of Knotted Surfaces* Jul. 2026
PCMI Experimental Math Lab

ACADEMIC SERVICE

Boston University Mathematics Association 2024–Present
Treasurer (2026–Present); Social Media Manager (2024–Present); Secretary (2025–2026)

SKILLS

Languages	Mandarin Chinese (native); English (fluent)
Programming	Magma, SageMath, JavaScript
Typesetting	L ^A T _E X